

Exp. No: 01**Date:****CLASSES AND OBJECTS**

PROGRAM :

```
import java.util.*;

class Box{

    double w,h,d;

    Box(){

        w=2;

        h=2;

        d=2;

    }

    Box(double a){

        w=a;

        h=2;

        d=3;

    }

    Box(double a,double b){

        w=a;

        h=b;

        d=3;

    }

    Box(double a,double b,double c){

        w=a;

        h=b;

        d=c;

    }

    double Volume(){

        return w*h*d;

    }

}
```

```
}  
  
class Vol_Box{  
  
    public static void main(String args[]) {  
  
        Scanner s=new Scanner(System.in);  
  
        System.out.println("\t Classes and Objects");  
  
        double a1,b1,b2,c1,c2,c3;  
  
        System.out.println("Enter One value :");  
  
        a1=s.nextDouble();  
  
        System.out.println("Enter Two values :");  
  
        b1=s.nextDouble();  
  
        b2=s.nextDouble();  
  
        System.out.println("Enter Three values :");  
  
        c1=s.nextDouble();  
  
        c2=s.nextDouble();  
  
        c3=s.nextDouble();  
  
        Box box=new Box();  
  
        Box box1=new Box(a1);  
  
        Box box2=new Box(b1,b2);  
  
        Box box3=new Box(c1,c2,c3);  
  
        System.out.println("The Volume of Box1 is :"+box.Volume());  
  
        System.out.println("The Volume of Box2 is :"+box1.Volume());  
  
        System.out.println("The Volume of Box3 is :"+box2.Volume());  
  
        System.out.println("The Volume of Box4 is :"+box3.Volume());  
  
    }  
  
}
```

OUTPUT :

```
ragupathi@ragu-2k05:~$ java Vol_Box
Classes and Objects
Enter One value :
2
Enter Two values :
2
3
Enter Three values :
4
5
6
The Volume of Box1 is :8.0
The Volume of Box2 is :12.0
The Volume of Box3 is :18.0
The Volume of Box4 is :120.0
```


Exp. No : 02**Date:****IMPLEMENTING CLASSES**

PROGRAM :

```
import java.util.*;

class Area_Shapes{

    double pi=3.14;

    void area(int a){

        System.out.println("\nThe Area of the Square is :"+a*a);

    }

    void area(int a,int b){

        System.out.println("The Area of the Rectangle is :"+a*b);

    }

    double area(double r){

        return pi*r*r;

    }

}

class Shapes{

    public static void main(String args[]){

        System.out.println("\t Implementing Classes");

        Scanner s=new Scanner(System.in);

        double c,area_cir;

        int sq,l,b;

        Area_Shapes ar=new Area_Shapes();

        System.out.print("\nEnter the Side for Square :");

        sq=s.nextInt();

        System.out.print("Enter the Length of the Rectangle :");

        l=s.nextInt();

        System.out.print("Enter the Breadth of the Rectangle :");

        b=s.nextInt();
```

```
System.out.print("Enter the Radius of the Circle :");  
c=s.nextDouble();  
ar.area(sq);  
ar.area(l,b);  
area_cir=ar.area(c);  
System.out.println("The Area of the Circle is :"+area_cir);  
}  
}
```

OUTPUT :

```
ragupathi@ragu-2k05:~$ java Shapes
    Implementing Classes

Enter the Side for Square :2
Enter the Length of the Rectangle :3
Enter the Breadth of the Rectangle :4
Enter the Radius of the Circle :4

The Area of the Square is :4
The Area of the Rectangle is :12
The Area of the Circle is :50.24
```


Exp. No: 03**Date:****STRINGS**

PROGRAM :

```

import java.util.*;

class String_func{

    public static void main(String args[]) {

        Scanner s=new Scanner(System.in);

        System.out.print("Enter the String :");

        String str=s.nextLine();

        System.out.print("Enter the string to check the given string is starts with it :");

        String a1=s.nextLine();

        System.out.print("Enter the string to check the given string is Ends with it :");

        String a2=s.nextLine();

        System.out.print("Enter the string to find position :");

        String p1=s.nextLine();

        System.out.print("Enter the position to find character :");

        int p2=s.nextInt();

        s.nextLine();

        System.out.println("\n");

        System.out.println("The length of the string is :"+str.length());

        System.out.println("The Upper case is :"+str.toUpperCase());

        System.out.println("The Lower case is :"+str.toLowerCase());

        System.out.println("The string is starts with the \""+a1+"\"
is :"+str.startsWith(a1));

        System.out.println("The string is ends with the \""+a2+"\"
is :"+str.endsWith(a2));

        System.out.println(p1+" position is :"+str.indexOf(p1));

        System.out.println("The character in "+p2+" is :"+str.charAt(p2));

        System.out.println("The substring is :"+str.substring(1,7));
    }
}

```

```
// String Buffer

System.out.println("\n----- StringBuffer Functions -----");

StringBuffer str2=new StringBuffer(str);

System.out.print("Enter the String to append :");

String ap=s.nextLine();

str2.append(ap);

System.out.println("After append :"+str2);

System.out.println("\n");

System.out.print("Enter the String to Insert :");

String i1=s.nextLine();

System.out.print("Enter the position to Insert :");

int i2=s.nextInt();

str2.insert(i2,i1);

System.out.println("After Insert :"+str2);

System.out.println("\n");

System.out.print("Enter the position to delete a character :");

int d1=s.nextInt();

str2.deleteCharAt(d1);

System.out.println("After delete : "+str2);

System.out.println("\n");

System.out.println("Reverse string :"+str2.reverse());

}

}
```

OUTPUT :

```
ragupathi@ragu-2k05:~$ java String_func
Enter the String :Hi Iam Ragupathi
Enter the string to check the given string is starts with it :H
Enter the string to check the given string is Ends with it :r
Enter the string to find position :R
Enter the position to find character :3

The length of the string is :16
The Upper case is :HI IAM RAGUPATHI
The Lower case is :hi iam ragupathi
The string is starts with the "H" is :true
The string is ends with the "r" is :false
R position is :7
The character in 3 is :I
The substring is :i Iam

----- StringBuffer Functions -----
Enter the String to append : stud
After append :Hi Iam Ragupathi stud

Enter the String to Insert : Hello
Enter the position to Insert :2
After Insert :Hi Hello Iam Ragupathi stud

Enter the position to delete a character :1
After delete : H Hello Iam Ragupathi stud

Reverse string :duts ihtapugaR maI olleH H
```


Exp. No: 04**Date:****COLLECTION**

PROGRAM :

```

import java.util.*;

class Collection{

    public static void main(String args[]) {

        Scanner s=new Scanner(System.in);

        ArrayList<Integer> al=new ArrayList<>();

        System.err.println("Enter Five Numbers :");

        for(int i=1;i<=5;i++){

            System.out.print("Enter the Element "+i+" :");

            al.add(s.nextInt());

        }

        int sum=0;

        for (int a:al){

            sum+=a;

        }

        System.out.println("The Sum of the Elements is :"+sum);

        LinkedList<String> l=new LinkedList<>();

        HashSet<String> hs=new HashSet<>();

        TreeSet<String> ts=new TreeSet<>();

        System.out.println("\nEnter Five Letters :");

        s.nextLine();

        for(int i=1;i<=5;i++){

            System.out.print("Enter the Letter "+i+" :");

            String str=s.nextLine();

            l.add(str);

            hs.add(str);

            ts.add(str);

        }

    }

}

```

```
    }  
    System.out.println("\nThe Linked List Elements are :"+l);  
    System.out.print("Enter the Element to remove :");  
    String s1=s.nextLine();  
    System.out.print("Enter the Position to remove :");  
    int s2=s.nextInt();  
    System.out.println("Before Remove :"+l);  
    l.remove(s1);  
    l.remove(s2);  
    System.out.println("After Remove :"+l);  
    System.out.println("\nThe HashSet(Unordered) elements are :"+hs);  
    System.out.println("\nThe TreeSet(Ordered) Elements are :"+ts);  
    }  
}
```

OUTPUT :

```
ragupathi@ragu-2k05:~$ java Collection
Enter Five Numbers :
Enter the Element 1 :2
Enter the Element 2 :3
Enter the Element 3 :4
Enter the Element 4 :5
Enter the Element 5 :6
The Sum of the Elements is :20

Enter Five Letters :
Enter the Letter 1 :A
Enter the Letter 2 :S
Enter the Letter 3 :D
Enter the Letter 4 :F
Enter the Letter 5 :G

The Linked List Elements are :[A, S, D, F, G]
Enter the Element to remove :S
Enter the Position to remove :2
Before Remove :[A, S, D, F, G]
After Remove :[A, D, G]

The HashSet(Unordered) elements are :[A, S, D, F, G]

The TreeSet(Ordered) Elements are :[A, D, F, G, S]
```


Exp. No: 05**Date:****DATE AND CALENDAR**

PROGRAM :

```
import java.util.*;

class Date_Time{

    public static void main(String[] args) {

        Scanner s=new Scanner(System.in);

        Calendar cal=Calendar.getInstance();

        Date dt=new Date();

        System.out.println("\t Date and Time" );

        System.out.println("\nThe Current Date,Time and Timezone in a System is :
        "+dt);

        System.out.println("The Current Time in Milli Second is :"+dt.getTime());

        String months[]=
        {"Jan","Feb","Mar","Apr","May","Jun","Jul","Aug","Oct","Sep","Nov","Dec"};

        System.out.print("\nDate : ");

        System.out.print(months[cal.get(Calendar.MONTH)]);

        System.out.print(" " + cal.get(Calendar.DATE));

        System.out.print(", " + cal.get(Calendar.YEAR));

        System.out.print("\nTime : ");

        System.out.print(cal.get(Calendar.HOUR));

        System.out.print(" : "+cal.get(Calendar.MINUTE));

        System.out.print(" : "+cal.get(Calendar.SECOND));

        System.out.println("\n");

        System.out.print("Enter the Hour :");

        int h=s.nextInt();

        System.out.print("Enter the Minutes :");
```

```
        int m=s.nextInt();  
        System.out.print("Enter the Seconds :");  
        int sc=s.nextInt();  
        cal.set(Calendar.HOUR,h);  
        cal.set(Calendar.MINUTE,m);  
        cal.set(Calendar.SECOND,sc);  
        System.out.print("\nUpdated Time :");  
        System.out.print(cal.get(Calendar.HOUR));  
        System.out.print(" : "+cal.get(Calendar.MINUTE));  
        System.out.print(" : "+cal.get(Calendar.SECOND)+"\n");  
    }  
}
```

OUTPUT :

```
ragupathi@ragu-2k05:~$ java Date_Time
    Date and Time

The Current Date,Time
and Timezone in a System is : Sat Sep 26 14:52:30 IST 2026
The Current Time in Milli Second is :1790414550901

Date : Oct 26, 2026
Time : 2 : 52 : 30

Enter the Hour :10
Enter the Minutes :20
Enter the Seconds :15

Updated Time :10 : 20 : 15
```


Exp. No: 06**Date:****PACKAGES**

PROGRAM :**Account.java :**

```
package Bank;

public class Account {

    String name;

    int balance;

    public void createAccount(String name, int initialAmount) {

        this.name = name;

        this.balance = initialAmount;

    }

    public int credit(int amount){

        balance+=amount;

        System.out.println("The Amount Credited Successfully");

        return balance;

    }

    public int withdraw(int amount){

        if(amount<=balance){

            balance-=amount;

            System.out.println("The Amount Withdrawn Successfully");

        }

        else{

            System.out.println("Your balance is Low");

        }

        return balance;

    }

    public void disp_balance(){

        System.out.println("Name :"+name);

    }

}
```

```

        System.out.println("Balance :"+balance);
    }
}

```

User.java :

```

import Bank.Account;

import java.util.*;

class User{

    public static void main(String[] args) {

        Scanner s= new Scanner(System.in);

        Account ac=new Account();

        System.out.print("\nEnter the Name :");

        String name=s.nextLine();

        System.out.print("\nEnter the Initial Amount :");

        int amount=s.nextInt();

        s.nextLine();

        ac.createAccount(name, amount);

        System.out.println("""

            1)Credit

            2)withdraw

            3)Balance

            4)Exit

            """);

        while(true){

            System.out.print("\nEnter What you want to do :");

            String msg=s.nextLine().toLowerCase();

            if("credit".equals(msg)){

```

```
        System.out.print("\nEnter the amount :");
        int amt=s.nextInt();
        ac.credit(amt);
        s.nextLine();
    }
    else if("withdraw".equals(msg)){
        System.out.print("\nEnter the amount :");
        int amt=s.nextInt();
        ac.withdraw(amt);
        s.nextLine();
    }
    else if("balance".equals(msg)){
        ac.disp_balance();
    }
    else if("exit".equals(msg)){
        break;
    }
    else{
        System.out.println("Invalid Action !");
    }
}
}
```

OUTPUT :

```
ragupathi@ragu-2k05:~$ java User
Enter the Name :Ragupathi
Enter the Initial Amount :1500
1)Credit
2)withdraw
3)Balance
4)Exit

Enter What you want to do :credit

Enter the amount :1000
The Amount Credited Successfully

Enter What you want to do :withdraw

Enter the amount :500
The Amount Withdrawn Successfully

Enter What you want to do :balance
Name :Ragupathi
Balance :2000

Enter What you want to do :exit
```


Exp. No: 07**Date:****EXCEPTION HANDLING**

PROGRAM :

```
import java.util.*;

class Exception_Hand {

    public static void main(String[] args) {

        int a = args.length;

        Scanner s = new Scanner(System.in);

        System.out.println("Value of a is : " + a);

        try {

            if (a == 0) {

                int b = 24;

                a = b / (b - b);

            }

            if (a == 1) {

                int A[] = {};

                A[16] = 20;

            }

            if (a == 2) {

                System.out.print("Enter Array size: ");

                int n = s.nextInt();

                int B[] = new int[n];

                B[17] = 15;

            }

        } catch (ArithmeticException e) {

            System.out.println("Do not use zero as a divisor. " + e);

        } catch (ArrayIndexOutOfBoundsException e) {

            System.out.println("Array size exceeds. " + e);

        }

    }

}
```

```
        catch (NegativeArraySizeException e) {  
            System.out.println("Array size cannot be negative value. " + e);  
        } finally {  
            System.out.println("Exceptions Completed.");  
        }  
    }  
}
```

OUTPUT :

```
ragupathi@ragu-2k05:~$ java Exception_Hand
Value of a is : 0
Do not use zero as a divisor. java.lang.ArithmeticException: / by zero
Exceptions Completed.
ragupathi@ragu-2k05:~$ java Exception_Hand 3
Value of a is : 1
Array size exceeds. java.lang.ArrayIndexOutOfBoundsException: Index 16 out of bounds for length 0
Exceptions Completed.
ragupathi@ragu-2k05:~$ java Exception_Hand 3 4
Value of a is : 2
Enter Array size: -5
Array size cannot be negative value. java.lang.NegativeArraySizeException: -5
Exceptions Completed.
```


Exp. No: 08**Date:****THREADS**

PROGRAM :

```
class Q {  
    int n;  
  
    boolean valueSet = false;  
  
    synchronized int get() {  
        if (!valueSet)  
            try {  
                wait();  
            } catch (InterruptedException e) {  
                System.out.println("InterruptedException caught");  
            }  
        System.out.println("Got: " + n);  
        valueSet = false;  
        notify();  
        return n;  
    }  
  
    synchronized void put(int n) {  
        if (valueSet)  
            try {  
                wait();  
            } catch (InterruptedException e) {  
                System.out.println("InterruptedException caught");  
            }  
        this.n = n;  
        valueSet = true;  
        System.out.println("Put: " + n);  
        notify();  
    }  
}
```

```
    }  
}  
  
class Producer implements Runnable {  
    Q q;  
    Producer(Q q) {  
        this.q = q;  
        new Thread(this, "Producer").start();  
    }  
    public void run() {  
        int i = 1;  
        while (i <= 10) {  
            q.put(i++);  
        }  
    }  
}  
  
class Consumer implements Runnable {  
    Q q;  
    Consumer(Q q) {  
        this.q = q;  
        new Thread(this, "Consumer").start();  
    }  
    public void run() {  
        while (true) {  
            q.get();  
        }  
    }  
}
```

```
class PC {  
    public static void main(String args[]) {  
        System.out.println("Press Control-C to stop.");  
        Q q = new Q();  
        new Producer(q);  
        new Consumer(q);  
    }  
}
```

OUTPUT :

```
ragupathi@ragu-2k05:~$ java PC
Press Control-C to stop.
Put: 1
Got: 1
Put: 2
Got: 2
Put: 3
Got: 3
Put: 4
Got: 4
Put: 5
Got: 5
Put: 6
Got: 6
Put: 7
Got: 7
Put: 8
Got: 8
Put: 9
Got: 9
Put: 10
Got: 10
```


Exp. No: 09**Date:****JDBC**

PROGRAM :

```
import java.sql.*;

public class Student {

    public static void main(String args[]) {

        try {

            // Load MySQL JDBC Driver

            Class.forName("com.mysql.cj.jdbc.Driver");

            // Database connection

            String url = "jdbc:mysql://localhost:3306/ragu";

            String user = "root";

            String password = "ragu@123";

            Connection con = DriverManager.getConnection(url, user, password);

            System.out.println("Created DB Connection...");

            Statement stmt = con.createStatement();

            // Execute Query

            ResultSet rs = stmt.executeQuery("SELECT Id, name, class FROM student");

            // Print Column Names

            ResultSetMetaData rsmd = rs.getMetaData();

            int count = rsmd.getColumnCount();

            for (int i = 1; i <= count; i++) {

                System.out.print(rsmd.getColumnName(i) + "\t");

            }

            System.out.println();

        }

    }

}
```

```
// Print Records

while (rs.next()) {

    System.out.print(rs.getInt("id") + "\t");

    System.out.print(rs.getString("name") + "\t");

    System.out.print(rs.getString("class") + "\t");

    System.out.println();

}

// Close Resources

rs.close();

stmt.close();

con.close();

} catch (ClassNotFoundException e) {

    System.out.println("JDBC Driver Not Found!");

    e.printStackTrace();

} catch (SQLException e) {

    System.out.println("Database Connection Error!");

    e.printStackTrace();

}

}
```

OUTPUT :

```
ragupathi@ragu-2k05:~$ java -cp ".:lib/mysql-connector.jar" Student
Created DB Connection...
Id      name      class
1       Ragu      12 th
2       Jeffy     11 th
3       Walter    10 th
```


Exp. No: 10**Date:****APPLETS**

PROGRAM :

```
import java.awt.*;

import java.applet.*;

/*

<applet code="SimpleBanner" width="700" height="50"> </applet>

*/

public class SimpleBanner extends Applet implements Runnable {

    String msg = " Accept Everything Nothing is Permanent.";

    Thread t = null;

    int state;

    boolean stopFlag;

    // Set colors and initialize thread

    public void init() {

        setBackground(Color.blue);

        setForeground(Color.white);

        setFont(new Font("Dialog", Font.PLAIN, 50));

    }

    // Start thread

    public void start() {

        t = new Thread(this);

        stopFlag = false;

        t.start();

    }

    // Entry point for the thread that runs the banner

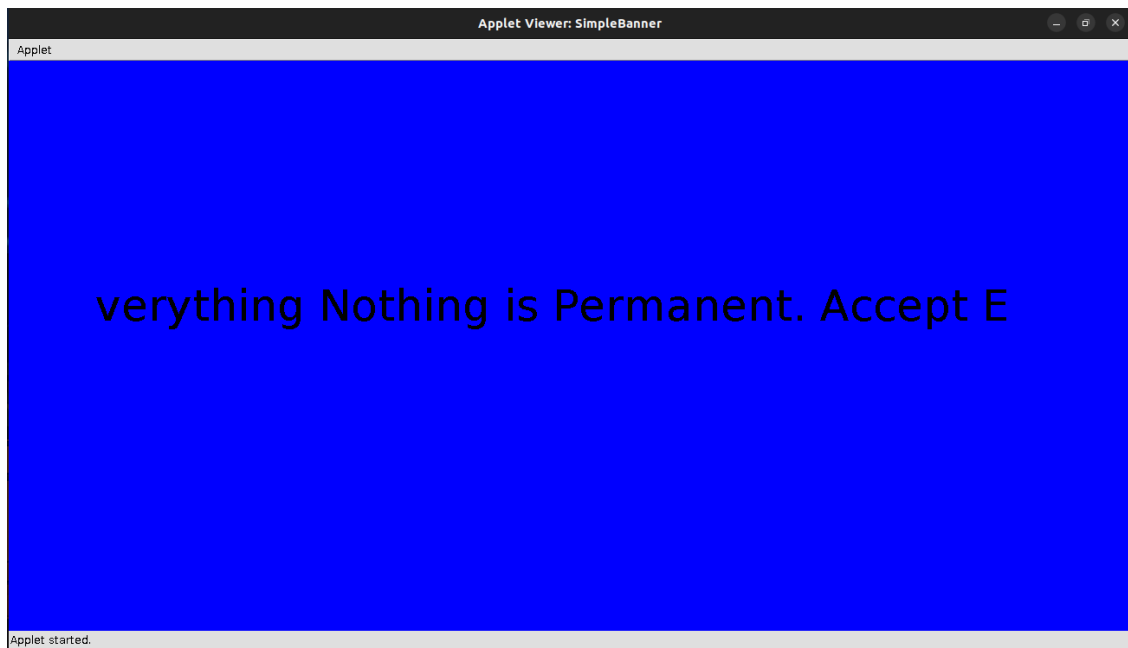
    public void run() {

        char ch;
```

```
// Display banner
for (;;) {
    try {
        repaint();
        Thread.sleep(250);
        ch = msg.charAt(0);
        msg = msg.substring(1, msg.length());
        msg += ch;
        if (stopFlag)
            break;
    } catch (InterruptedException e) {
    }
}

// Pause the banner
public void stop() {
    stopFlag = true;
    t = null;
}

// Display the banner
public void paint(Graphics g) {
    g.drawString(msg, 100, 300);
}
}
```

OUTPUT :

Exp. No: 11**Date:****EVENT HANDLING**

PROGRAM :

```
import java.applet.*;

import java.awt.*;

import java.awt.event.*;

/*

<applet code="Event_Hand" width="1200" height="1200">

</applet>

*/

public class Event_Hand extends Applet implements ActionListener {

    TextField a, b;

    double sum;

    Button ab;

    public void init() {

        Label value1 = new Label("First Number : ", Label.RIGHT);

        Label value2 = new Label("Second Number : ", Label.RIGHT);

        a = new TextField(10);

        b = new TextField(10);

        ab = new Button("Add");

        add(value1);

        add(a);

        add(value2);

        add(b);

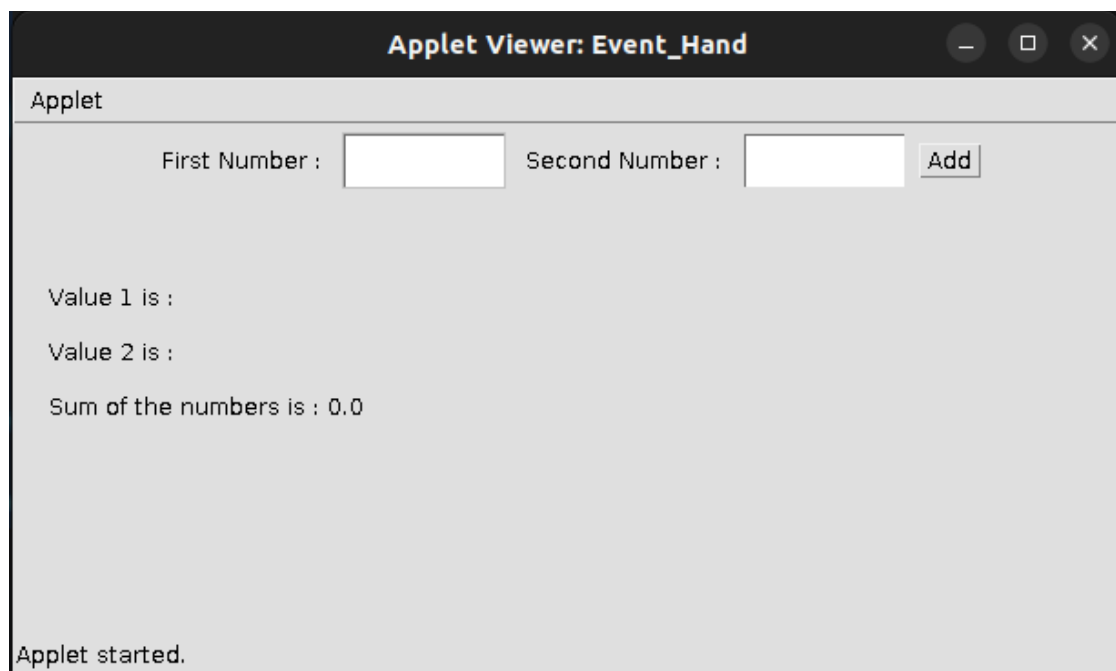
        add(ab);

        ab.addActionListener(this);

    }

    public void actionPerformed(ActionEvent ae) {
```

```
if (ae.getSource() == ab) {  
    try {  
        int num1 = Integer.parseInt(a.getText());  
        int num2 = Integer.parseInt(b.getText());  
        sum = num1 + num2;  
    } catch (NumberFormatException e) {  
        sum = 0;  
    }  
    repaint();  
}  
  
public void paint(Graphics g) {  
    g.drawString("Value 1 is : " + a.getText(), 20, 100);  
    g.drawString("Value 2 is : " + b.getText(), 20, 130);  
    g.drawString("Sum of the numbers is : " + sum, 20, 160);  
}  
}
```

OUTPUT :

Exp. No: 12**Date:****SIMPLE WEB APPLICATIONS**

PROGRAM :**Index.html :**

```
<html>

<head>

    <title>Voting Eligibility Check</title>

</head>

<body>

<h2>Check Voting Eligibility</h2>

<form action="vote" method="post">

    <label for="name">Name:</label>

    <input type="text" id="name" name="name" required>

    <br><br>

    <label for="age">Age:</label>

    <input type="number" id="age" name="age" required>

    <br><br>

    <input type="submit" value="Check Eligibility">

</form>

</body>

</html>
```

vote.java :

```
import java.io.IOException;

import java.io.PrintWriter;

import javax.servlet.ServletException;

import javax.servlet.annotation.WebServlet;

import javax.servlet.http.HttpServlet;

import javax.servlet.http.HttpServletRequest;
```

```
import javax.servlet.http.HttpServletResponse;

@WebServlet("/vote")

public class vote extends HttpServlet {

    @Override

    protected void doPost(HttpServletRequest request,

        HttpServletResponse response)

        throws ServletException, IOException {

        String name = request.getParameter("name");

        int age = Integer.parseInt(request.getParameter("age"));

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        out.println("<html><body>");

        if (age < 18) {

            out.println("<h2>" + name + " is not eligible to vote.</h2>");

        } else {

            out.println("<h2>" + name + " is eligible to vote.</h2>");

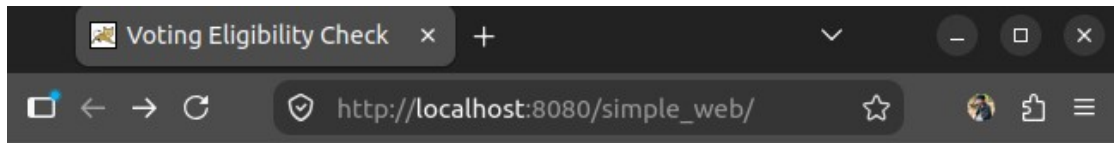
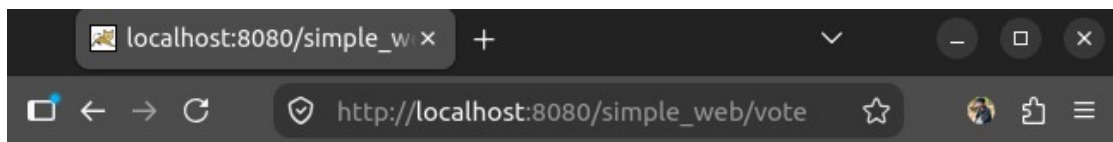
        }

        out.println("</body></html>");

        out.close();

    }

}
```

OUTPUT :**Check Voting Eligibility**Name: Age: **Ragupathi is eligible to vote.**

Exp. No: 13**Date:****USING SESSIONS AND COOKIES**

PROGRAM :**Index.html :**

```
<html>

<head>

    <title>Login Page</title>

</head>

<body>

<h2>Login</h2>

<form action="LoginServlet" method="POST">

    Name:

    <input type="text" name="username">

    <br><br>

    <input type="submit" value="Submit">

</form>

</body>

</html>
```

LoginServlet.java :

```
import java.io.*;

import javax.servlet.*;

import javax.servlet.http.*;

import javax.servlet.annotation.WebServlet;

@WebServlet("/LoginServlet")

public class LoginServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request,

                           HttpServletResponse response)
```

```

        throws ServletException, IOException {
    response.setContentType("text/html");

    String username = request.getParameter("username");

    // Create Session

    HttpSession session = request.getSession();
    session.setAttribute("username", username);

    // Create Cookie

    Cookie userCookie = new Cookie("username", username);
    userCookie.setMaxAge(60 * 60 * 24); // 1 day
    response.addCookie(userCookie);
    response.sendRedirect("WelcomeServlet");
}
}

```

WelcomeServlet.java :

```

import java.io.*;

import javax.servlet.*;

import javax.servlet.http.*;

import javax.servlet.annotation.WebServlet;

@WebServlet("/WelcomeServlet")

public class WelcomeServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request,

        HttpServletResponse response)

        throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();
    }
}

```

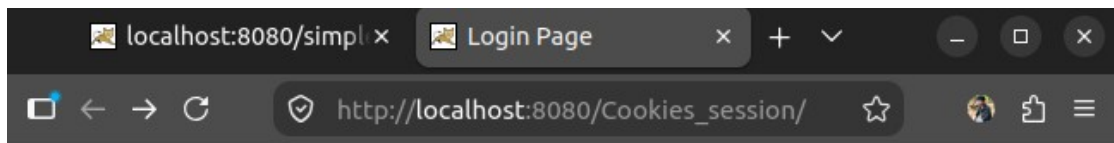
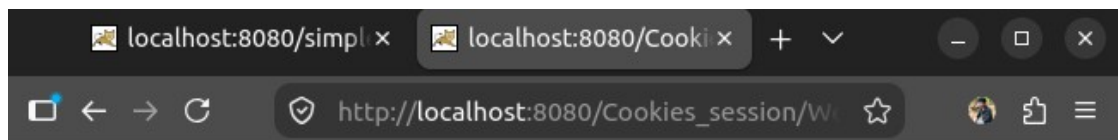
```
HttpSession session = request.getSession(false);

String username = null;

if (session != null) {
    username = (String) session.getAttribute("username");
}

if (username == null) {
    Cookie[] cookies = request.getCookies();
    if (cookies != null) {
        for (Cookie cookie : cookies) {
            if (cookie.getName().equals("username")) {
                username = cookie.getValue();
                break;
            }
        }
    }
}

if (username != null) {
    out.println("<h2>Welcome, " + username + "!</h2>");
} else {
    out.println("<h2>No user found. Please log in.</h2>");
    out.println("<a href='index.html'>Login</a>");
}
}
```

OUTPUT :**Login**Name: **Welcome, Ragupathi!**

Exp. No: 14**Date:****FORWARDING REQUESTS AND REDIRECTING
RESPONSES**

PROGRAM :**Homepage.html :**

```
<html>

<head>

    <title>Age Verification</title>

</head>

<body>

<h2>Enter Your Details</h2>

<form action="NewServlet" method="post">

    Name:

    <input type="text" name="username" required><br><br>

    Age:

    <input type="number" name="age" required><br><br>

    <input type="submit" value="Submit">

</form>

</body>

</html>
```

Error.html :

```
<html>

<head> <title>Error</title> </head>

<body>

<h2>You are not eligible to vote.</h2>

<p>You must be 18 years or older to vote.</p>

<a href="homepage.html">Go Back</a>

</body>

</html>
```

NewServlet.java :

```
import java.io.IOException;

import javax.servlet.RequestDispatcher;

import javax.servlet.ServletException;

import javax.servlet.annotation.WebServlet;

import javax.servlet.http.HttpServlet;

import javax.servlet.http.HttpServletRequest;

import javax.servlet.http.HttpServletResponse;


@WebServlet("/NewServlet")

public class NewServlet extends HttpServlet {

    protected void processRequest(HttpServletRequest request,

        HttpServletResponse response)

        throws ServletException, IOException {

        String username = request.getParameter("username");

        int age = Integer.parseInt(request.getParameter("age"));

        if (age >= 18) {

            request.setAttribute("username", username);

            RequestDispatcher rd =

                request.getRequestDispatcher("Welcome");

            rd.forward(request, response);

        } else {

            response.sendRedirect("error.html");

        }

    }

    @Override
```

```

protected void doGet(HttpServletRequest request,
                    HttpServletResponse response)
                    throws ServletException, IOException {
    processRequest(request, response);
}

@Override

protected void doPost(HttpServletRequest request,
                    HttpServletResponse response)
                    throws ServletException, IOException {
    processRequest(request, response);
}
}

```

Welcome.java :

```

import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/Welcome")

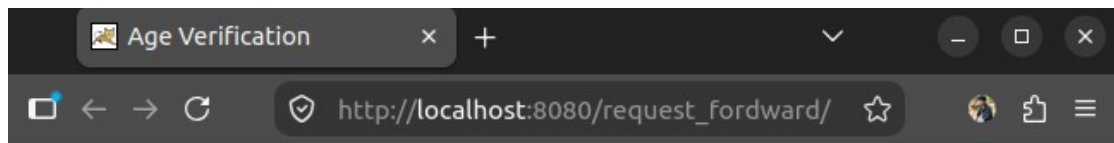
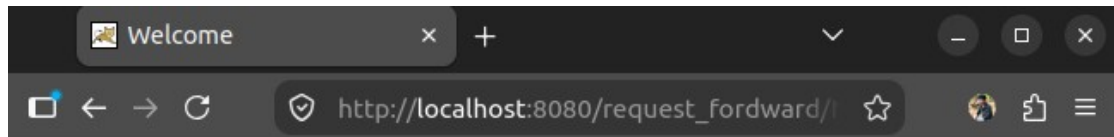
public class Welcome extends HttpServlet {
    protected void processRequest(HttpServletRequest request,
                                HttpServletResponse response)
                                throws ServletException, IOException {

```

```
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        String username = (String) request.getAttribute("username");
        out.println("<!DOCTYPE html>");
        out.println("<html>");
        out.println("<head>");
        out.println("<title>Welcome</title>");
        out.println("</head>");
        out.println("<body>");
        out.println("<h2>Welcome " + username + "!</h2>");
        out.println("<p>You are eligible to vote.</p>");
        out.println("</body>");
        out.println("</html>");
    }

    @Override
    protected void doGet(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {
        processRequest(request, response);
    }

    @Override
    protected void doPost(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {
        processRequest(request, response);
    }
}
```

OUTPUT :**Enter Your Details**Name: Age: **Welcome Ragu!**

You are eligible to vote.

Exp. No: 15**Date:****WEB APPLICATIONS USING DATABASE**

PROGRAM :**Index.html :**

```
<html>

<head>

    <title>Login Page</title>

</head>

<body>

<form action="LoginServlet" method="post">

    <h2>Login Form</h2>

    Username:

    <input type="text" name="username" required><br><br>

    Password:

    <input type="password" name="password" required><br><br>

    <input type="submit" value="Login">

</form>

</body>

</html>
```

DatabaseConnection.java :

```
import java.sql.Connection;

import java.sql.DriverManager;

public class DatabaseConnection {

    static Connection con;

    public static Connection initializeDatabase() {

        try {

            Class.forName("com.mysql.cj.jdbc.Driver");
```

```

        con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/ragu",
            "root",
            "ragu@123");
    } catch (Exception e) {
        e.printStackTrace();
    }
    return con;
}
}

```

LoginServlet.java :

```

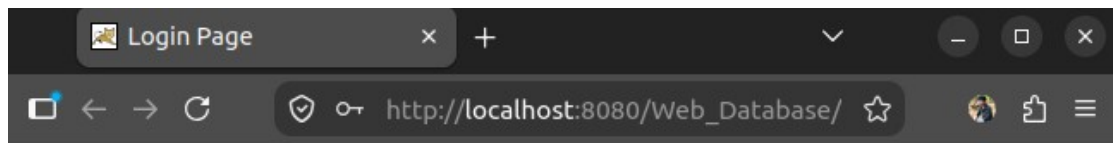
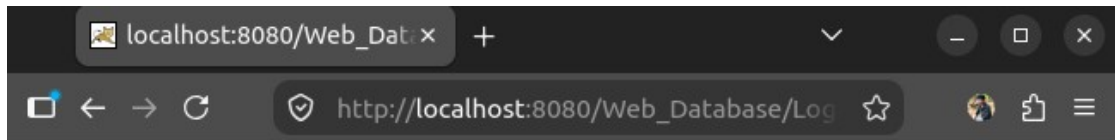
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/LoginServlet")
public class LoginServlet extends HttpServlet {
    @Override

```

```
protected void doPost(HttpServletRequest request,
    HttpServletResponse response)
    throws ServletException, IOException {
    response.setContentType("text/html");
    PrintWriter out = response.getWriter();
    String username = request.getParameter("username");
    String password = request.getParameter("password");
    try {
        Connection con = DatabaseConnection.initializeDatabase();
        String sql = "SELECT * FROM stud WHERE user=? AND pass=?";
        PreparedStatement ps = con.prepareStatement(sql);
        ps.setString(1, username);
        ps.setString(2, password);
        ResultSet rs = ps.executeQuery();
        if (rs.next()) {
            out.println("<html>");
            out.println("<body>");
            out.println("<h2>Login Successful</h2>");
            out.println("<h3>Welcome " + username + "</h3>");
            out.println("</body>");
            out.println("</html>");
        } else {
            out.println("<html>");
            out.println("<body>");
            out.println("<h2>Invalid Username or Password</h2>");
            out.println("</body>");
            out.println("</html>");
        }
    }
}
```

```
        }  
        con.close();  
    } catch (Exception e) {  
        out.println("<h2>Database Error</h2>");  
        e.printStackTrace(out);  
    }  
}  
}
```

OUTPUT :**Login Form**Username: Password: **Login Successful****Welcome ragu**

Exp. No: 16**Date:****DEVELOPING SIMPLE BEANS**

PROGRAM :**Student.java :**

```
package stud;

public class Student {

    private String name;

    private int id;

    public Student() {

    }

    public Student(String name, int id) {

        this.name = name;

        this.id = id;

    }

    public String getName() {

        return name;

    }

    public void setName(String name) {

        this.name = name;

    }

    public int getId() {

        return id;

    }

    public void setId(int id) {

        this.id = id;

    }

}
```

TestStudent.java :

```
import stud.Student;

public class TestStudent {

    public static void main(String[] args) {

        Student s1 = new Student();

        s1.setName("Ragu");

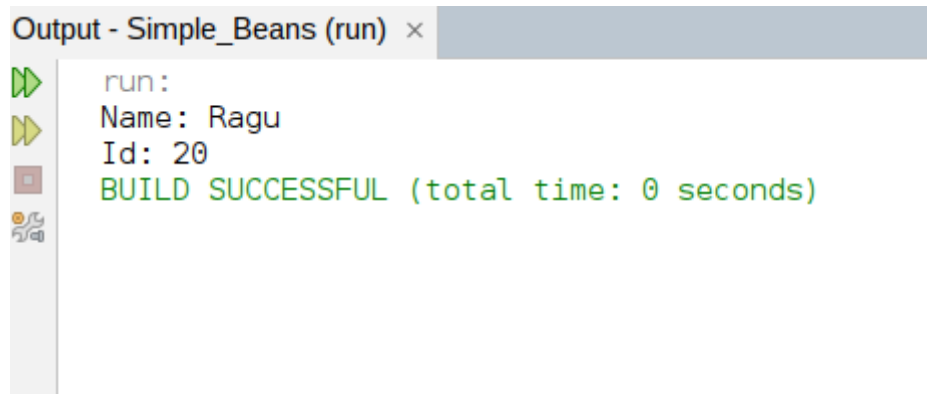
        s1.setId(20);

        System.out.println("Name: " + s1.getName());

        System.out.println("Id: " + s1.getId());

    }

}
```

OUTPUT :

```
run :
Name: Ragu
Id: 20
BUILD SUCCESSFUL (total time: 0 seconds)
```


Exp. No: 17**Date:****USE BEANS WITH JSP TAGS**

PROGRAM :**Index.jsp :**

```
<%@ page import="student.Student" %>

<html>

<head>

    <title>Student Details</title>

    <style>

        h1{

            color: blue;

        }

    </style>

</head>

<body>

<h1>Student Details</h1>

<%

    Student student = new Student();

    student.setName("Ragupathi");

    student.setId(21);

    request.setAttribute("student", student);

%>

<p>Name : ${student.name}</p>

<p>ID : ${student.id}</p>

</body>

</html>
```

Student.java :

```
package student;

public class Student {

    private String name;

    private int id;

    public Student() {

    }

    public String getName() {

        return name;

    }

    public void setName(String name) {

        this.name = name;

    }

    public int getId() {

        return id;

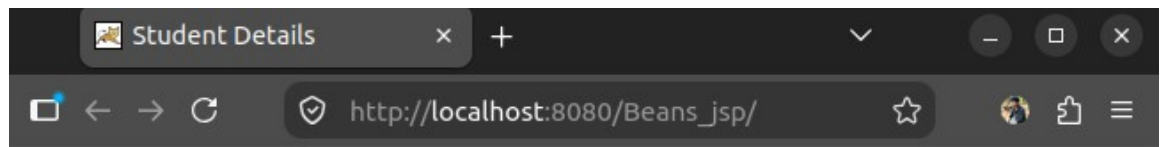
    }

    public void setId(int id) {

        this.id = id;

    }

}
```

OUTPUT :

Student Details

Name : Ragupathi

ID : 21

